

NICOLE FAVELA

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EDUCATION

The University of Texas at El Paso

Master of Science in Computer Science **GPA: 4.0/4.0**

The University of Texas at El Paso

Bachelor of Science in Computer Science with Minor in Mathematics **GPA: 3.82/4.0**

WORK EXPERIENCE

Full Stack Engineer Intern at Careforge AI

June 2025-Present

- Leveraged Amazon web services (**AWS**), **FastAPI**, **Langchain**, and **PGVector** to create and deploy therapy focused **RAG** chatbot using microservice architecture.
- Integrated **s3** storage and **lambda** triggered processing of knowledge base documents to provide an efficient and HIPAA compliant data pipeline.

Lead Software Engineer at Series - The first AI social network

April 2024-April 2025

- Leveraged Google Cloud Platform (**GCP**) services such as Storage Buckets, **Cloud Functions**, and the CLI to streamline cloud-based operations and automate workflows.
- Enhanced Vera AI chatbot's user experience and guidance capabilities by optimizing natural language understanding (NLU) increasing user satisfaction
- Mentored and onboarded new interns, fostering their growth through hands-on guidance and structured support, reducing onboarding time by 40%.

Interim CTO at Series - The first AI social network

Sept. - Oct. 2024

- Led the technical vision and architecture for Series core product, guiding interns across full stack technologies including **Node js**, **Firebase** and other **GCP** products.
- Oversaw cross-functional engineering efforts and implemented **Agile** practices and organization to accelerate product development and team efficiency and communication.
- Integrated intelligent features and optimized backend infrastructure that included data migration and database design.

PROJECTS:

Self-RAG Research Project

May 2026

- Designed and implemented a prompt-based approximation of the Self-RAG framework (Asai et al., 2023) using Llama 3.1 8B, achieving 21% Exact Match on PopQA — outperforming Vanilla RAG baseline across EM, F1, and ROUGE without any model fine-tuning.
- Built a Wikipedia retrieval corpus using FAISS vector search and Sentence-Transformers, and engineered a LLM self-assessment with cosine similarity scoring to address prompted model overconfidence.
- Conducted reflection token calibration analysis across 200 evaluation questions, producing quantitative evidence that selective retrieval reduces unnecessary retrieval by 3% while improving answer quality across all automatic metrics.

Pong Playing Atari Reinforcement Learning Agent

May 2025

- Developed a fully connected neural network using **PyTorch** and **Q-learning** to predict optimal paddle actions from Atari Pong frame coordinate data.
- Engineered data preprocessing methods for cleaning, transforming, and normalizing Gymnasium environment outputs into NumPy arrays for model input.
- Visualized agent performance and learning progression using **Matplotlib**, demonstrating convergence and behavior over time through training metrics.

EXTRACURRICULAR ACTIVITIES

Clubs & Honors: RTC (Rewriting the Code), Women Who Do Data (W2D2), and SWE (Society of Women Engineers) member, FOSS (Free and Open-Source Software) club member, HITEC Scholar, Phi Kappa Phi Honors Society member

Hackathons & Events: DEVCOM Cyber-Security hackathon, DEVCOM Dragon's Den Capture the Flag (2024)

SKILLS

- Frontend:** React.js, Next.js, Bootstrap, MUI
- Backend:** Flask, Java, FastAPI, Node.js, TypeScript, Next.js, Express.js, MongoDB, Postgres, Google Cloud Firestore, and MySQL, Pinecone vector database, PGVector
- Tools and Technologies:** Figma, Jira, Trello, n8n, Google Cloud CLI, AWS CLI, Swagger, OpenAI Embeddings, OpenAI API, PDFLoader, PyMuPdf, Cursor, Docker
- Cloud:** AWS s3, AWS lambda functions, AWS Secret Manager, AWS SAM, Google Cloud functions